

Scale-Up Implementation Plan — Construction-VLM XAI Evaluation

Version: 1.0 · Status: draft for review · Prerequisite: two-week Florence-2 pilot complete (163 samples, all six metrics ran end-to-end).

This document is the engineering specification for scaling the pilot into a full study. It is written to be implementable by an engineer or agent without prior context on the pilot. It contains no code, but it specifies every architecture, pipeline, data, and hyperparameter choice needed to build the system. Where a value is a deliberate decision rather than an arbitrary default, the rationale is given. Items marked [DECISION] require Professor Abdallah's sign-off before implementation; a recommended default is provided for each so work is not blocked.

0. Orientation: what exists today, and the core problem to fix

The pilot pipeline is an 11-script sequence: environment check → balanced sample selection → baseline inference → region standardization → the five per-sample metrics (descriptive accuracy, sparsity, stability, robustness) plus efficiency → bounded completeness → cross-metric rollup. It runs on Florence-2-base-ft against the `LouisChen15/ConstructionSite` Hugging Face dataset.

The four safety rules the pipeline evaluates:

- rule_1 — basic PPE present (hard hat / hi-vis vest) → class `ppe_violation`
- rule_2 — fall protection (harness/lanyard) at height → class `fall_hazard`
- rule_3 — edge protection (guardrail) at height/excavation edges → class `fall_hazard`
- rule_4 — worker inside excavator blind-spot/swing radius → class `struck_by_risk`
- (no violation on any rule → class `compliant`)

The single most important fact for this plan: on Florence-2, the safety answer is not produced by the model. Florence-2 only returns bounding boxes for phrases ("worker", "hard hat", ...). The compliant/violation judgment is then computed by hand-written geometric Python (per-worker coverage within a fixed pixel-distance threshold). Consequently, several XAI metrics partly measure the sensitivity of that geometric proxy rather than the model's own reasoning. The scale-up's central scientific goal is to make the metrics measure model faithfulness, which requires (a) fixing the proxy's known biases and (b) adding a model that makes the judgment itself (native-VQA). Everything below serves that goal.

1. Guiding principles (apply to every phase)

1. Freeze the pilot record. All new behavior lands behind configuration flags. The recorded 163-sample pilot numbers must remain reproducible byte-for-byte. Never mutate a metric's default behavior in place; add a new mode and make the old one the default until the new one is validated.
2. Every number traces to a results file. No figure or table value is ever hand-typed into a report; report artifacts are generated programmatically from the results CSVs at build time.
3. Parametrize everything that could change with scale. No sample count, class size, image resolution, grid dimension, or model ID may be hardcoded in analysis or plotting code. (The pilot shipped a bug where `n=163` and class sizes were literal strings; that class of bug must be designed out.)
4. Separate model-faithfulness signal from proxy-logic signal. Whenever a metric's result could be produced by the geometric proxy's own branching rather than the model, instrument for it and report the corrected number alongside

the raw one.

5. Report the metric's true target, never only the ranking heuristic's pick. For every region-based metric, report both the ranked top region and the rule's actual queried object.
6. Determinism where it matters, controlled stochasticity where it's the point. Unify decoding across metrics (see Phase 2). Use multiple seeds for anything sampling-based.

2. Phase 0 — Decisions to lock before build (with recommended defaults)

These four decisions change how much of Phases 2–5 gets built. Recommended defaults let Phase 1 proceed in parallel.

#	Decision	Recommended default	Consequence if changed
D1	Attribution method. Accept decoder→encoder cross-attention, or add a transformer-native gradient method (Integrated Gradients / DeepLIFT / a transformer LRP variant)	Add Integrated Gradients (IG) via Captum as a second, independent attribution stream, keeping cross-attention as the primary. Report agreement between the two.	If gradient methods are required, model must run in bf16 or fp32 for attribution (fp16 underflows gradients), and attribution compute grows 10–50×.
D2	Model scope. Florence-2 only, or add a native-VQA model	Add Qwen2.5-VL (3B and 7B) as native-VQA comparison; keep Florence-2 as the region-native baseline; one frontier model (e.g. GPT-4o) as benchmark-only reference.	Determines whether metrics measure model faithfulness (native-VQA) or proxy sensitivity (Florence-2 only).
D3	Class labeling. Keep mutually-exclusive priority collapse, or move to multi-label	Multi-label. An image can be ppe_violation AND struck_by_risk simultaneously.	Recovers the scarce struck_by_risk class (13 → ~70 across the full dataset) and enables multi-hazard completeness testing the proposal promised.
D4	Detection task token. Keep single-box <OPEN_VOCABULARY_DETECTION>, or move to exhaustive instance enumeration	Evaluate <DENSE_REGION_CAPTION> / <OD> on a validation subset first; adopt only if it measurably improves multi-worker recall. Do not assume any token "guarantees" full enumeration — measure it.	Multi-worker detection is a precondition for valid completeness and for rule_4 (proximity) correctness; but longer outputs raise decode cost and make beam-vs-greedy matter (see Phase 5 efficiency).

Minimum-n / statistics standard [DECISION, deferred to D2/D3 outcome]: set a minimum samples-per-class before a metric is reported as a stable estimate (recommend $n \geq 30$ per class per rule as a floor), and adopt a nonparametric significance test (Wilcoxon signed-rank, matching Prof. Abdallah's prior IDS work) plus bootstrap confidence intervals on every headline number.

3. Phase 1 — Foundational correctness fixes (highest leverage, lowest risk)

These are cheap, require no new models, and unblock six downstream chapters. Do these first. All land behind flags; default off until validated, then flipped on for the scale run.

3.1 Rule-aware region ranking (fixes the pilot's #1 issue)

- Problem: candidate explanation regions are ranked by pixel area, so the worker's body box outranks the small PPE object in ~96% of PPE samples; every masking/overlap metric then tests the wrong box.
- New ranking policy: rank candidate regions by a semantic-match-first key: (1) regions whose label matches the rule's queried object class rank above all others; (2) within a tier, rank by a real importance signal if available

(cross-attention mass inside the box — see Phase 4), else by area as a last resort.

- Object-class map: rule_1 → {hard hat, hi-vis vest}; rule_2 → {safety harness, lanyard}; rule_3 → {guardrail, edge barrier}; rule_4 → {excavator} (proximity partner) with worker as the subject.
- Config: region_ranking □ {area, rule_aware, attention}; default area (frozen pilot), scale run uses rule_aware.
- Validation gate: re-run descriptive accuracy and completeness; confirm the previously-measured 11.9-percentage-point "mask worker vs mask object" gap narrows substantially. Record before/after.

3.2 Multi-label classification + class-balanced sampling (fixes struck_by_risk scarcity)

- Problem: priority-collapse assigns each image exactly one class, discarding the 11 of 24 real struck_by_risk test-split images that also had a higher-priority violation. This is a labeling artifact, not a dataset property.
- Fix: classification returns the full set of violated classes per image (it is already computed internally, then discarded). Sampling becomes multi-label-aware: draw a target count per class such that every class reaches the minimum-n floor, allowing images to count toward multiple classes.
- Per-rule test assignment: an image is tested against every rule it is labeled for (not one). Compliant images are assigned rules via context-aware matching where feasible (see 3.3), else round-robin as a fallback.
- Config: labeling □ {priority, multilabel}; default priority, scale run uses multilabel.
- Expected effect: struck_by_risk usable pool rises from 13 to ~70 across the full 10k; enables genuine multi-hazard completeness tests.

3.3 Compliant-sample context matching (reduces vacuous tests)

- Problem: a compliant flat-ground image round-robin to rule_2 (harness/height) tests nothing — the hazard context doesn't exist.
- Fix: when assigning a rule to a compliant image, prefer a rule whose context is present in the scene (e.g. only assign rule_2/rule_3 to images with a height/edge context signal; only assign rule_4 where an excavator is present). Use dataset metadata fields where available; fall back to a lightweight grounding pre-check. Where no contextual match exists, mark the assignment context_absent so it can be reported separately rather than silently inflating baseline accuracy.
- Config: compliant_assignment □ {round_robin, context_matched}; default round_robin.

3.4 Worker-loss instrumentation baked into the metrics (de-contaminates 3 metrics)

- Problem: masking or perturbing the worker box can make the worker undetectable, which reroutes the proxy into a degenerate scene-level fallback and flips the answer for a reason unrelated to the safety concept. This inflates descriptive accuracy, robustness, and completeness.
- Fix: every metric that reruns inference after a mask/perturbation must record post-operation worker_boxes and emit a flip_due_to_worker_loss boolean. Report each metric twice: raw (all flips) and genuine (excluding worker-loss flips). This promotes the pilot's ad-hoc 159-rerun correction into standard metric output.
- Config: report_worker_loss_corrected = true for the scale run.

3.5 Prompt dictionary fix (rule_2 concept split)

- Problem: rule_2 groups "safety harness" and "fall-protection lanyard" as synonyms; they are two distinct physical objects, so a rewording test spuriously grounds different boxes.
- Fix: restructure the query dictionary so each distinct physical object has its own entry. True synonyms (guardrail / edge protection barrier) stay grouped; distinct objects (harness / lanyard) become separate concepts, each with its own compliance semantics. The rewording-robustness test only ever swaps validated synonyms.

- Deliverable: a reviewed query dictionary with, for each rule, an explicit {concept: [synonym, ...]} structure and a note on whether presence of any one concept, or all, satisfies the rule.

3.6 Unify decoding (removes the three-regime fragmentation)

- Problem: baseline/masking use beam search (num_beams=3), attribution uses greedy (num_beams=1), stability uses sampling — so metrics evaluate three different execution paths.
- Fix: adopt greedy decoding (num_beams=1) as the single canonical path for baseline, descriptive accuracy, sparsity, robustness, and the anchor of stability. Rationale: (a) it is the path attribution can be extracted from without beam-reordering bookkeeping; (b) efficiency profiling showed beam≈greedy cost for short grounding outputs, so nothing is lost. Stability's stochastic reruns remain sampling-based (that is the metric's point) but are compared against the greedy anchor.
- Config: decoding = greedy globally; stability.sampling_temperature retained as a separate parameter (see Phase 3).
- Caveat to record: if D4 adopts dense captioning (long outputs), re-evaluate beam-vs-greedy cost — the decode loop stops being free.

3.7 Hardcoded-assumption audit

- Sweep every analysis/plotting script for literal sample counts, class sizes, grid dimensions, patch-grid arithmetic (e.g. the 768→24 image-token math), and model IDs. Replace each with a value read from data or config. Add an automated check that fails CI if a known-magic literal reappears.

4. Phase 2 — Metric-validity fixes (make each metric measure what it claims)

Each of the six metrics needs an explicit control for an artifact the tabular original never faced. All additive; none replaces the pilot metric.

4.1 Descriptive Accuracy

- Add text-token ablation (the missing multimodal half the proposal promised): in addition to masking the visual region, remove the key query token(s) and re-measure the answer flip. Report visual-ablation and text-ablation flip rates separately, never merged.
- Add a graded score: expose a continuous proxy signal (fraction of workers covered / minimum distance ratio) so masking effects can be measured as magnitude, not just a boolean flip — this recovers statistical power and smooths the non-monotonicity anomaly.
- Add a monotonic-masking invariant: flag any sample where masking more regions un-flips an answer (the fallback-interaction signature) for audit.
- Mask-mode sensitivity: run a subset under black, blur, and inpaint masking; confirm flip rate is robust to mask mode (black patches are out-of-distribution artifacts).

4.2 Visual Sparsity

- Replace/augment the top-k-of-576 concentration measure with a size-invariant metric: attention mass inside the grounded box ÷ the box's cell footprint (an excavator covering 40% of the frame should have its attention judged against 40% of cells, not against all 576). This removes the −0.70 size confound that makes large-object rules look artificially unfocused.
- Fix and document one normalization convention (mass ratio on raw softmax attention; threshold count on min-max-normalized) and keep it consistent across the run.

- Reduce over-averaging: evaluate head/layer selection (informative heads, or last decoder layer) and per-generated-token attribution instead of averaging over all layers/heads/steps; pick the variant with the best signal on a validation subset and freeze it.

4.3 Stability

- Report both overlap scores by rule: the ranked top-region overlap and the rule's true object-box overlap.
- Add a size-invariant stability measure (normalized centroid-distance and/or Dice) alongside IoU, so small PPE boxes aren't penalized purely by IoU's size bias.
- Add an object-presence-stability metric: fraction of reruns in which the object is detected at all. (The pilot silently dropped samples where the box appeared/disappeared across reruns — the worst instability — by returning NaN; that must count as instability, not vanish.)
- Temperature: replace the single arbitrary temperature=0.7 with a sweep (e.g. {0.3, 0.5, 0.7, 1.0}) and report a stability-vs-temperature curve. Increase reruns per sample from 3 to ≥ 10 for smoother per-sample estimates.

4.4 Robustness

- Severity sweeps (dose-response): each perturbation gets multiple magnitudes, not one fixed point. Recommended grids: Gaussian blur kernel {4, 8, 16}; gamma {1.5, 2.5, 4.0}; occlusion area-fraction {0.1, 0.2, 0.35} at both centered and random locations; contrast factor {0.15, 0.3, 0.5}. Report answer-level and explanation-level change as curves vs severity.
- Two-number reporting (keep from pilot): answer-level robustness and explanation-level (box-drift) robustness, never blended.
- Verify the centerpiece finding under a size control: the pilot's headline "28.3% of stable answers hide a drifted explanation" uses IoU, which is size-biased — bucket by object-box area and confirm the drift survives within size bands; add a size-invariant drift measure. This must be checked before the paper leans on the number.
- Report drift and box-disappearance jointly (don't let vanished boxes drop out of the drift denominator).
- Add attention-heatmap drift (from Phase 4) as a second explanation-level signal beyond box IoU.
- Confidence proxy: retain only as an explicitly-labeled uncalibrated diagnostic; never report as a correctness/trust signal (it rose under the most disruptive perturbations).
- Occlusion: separate targeted occlusion (cover the actual object box) from random-location occlusion, since the pilot's centered square conflated "covered the object" with "disrupted whole-scene grounding."

4.5 Bounded Completeness

- Bake the worker-loss correction (3.4) into the headline number.
- Report completeness with an explicit note that it shares construction with descriptive accuracy (they are not independent confirmations).
- Multi-hazard corner cases (the proposal's core promise): with multi-label data (3.2), deliberately stratify and oversample images that violate two or more rules; test whether the explanation is complete enough to capture both hazards (e.g. missing hard hat AND unprotected edge). Report multi-hazard completeness as its own subsection with adequate n.
- Optionally extend beyond top-1/top-2 ablation toward a higher-order necessity check (a region necessary only in combination).

4.6 Efficiency

- Instrument real per-sample timing into every inference-bearing script (retire the 15-sample calibration).

- Report GPU-compute time separately from end-to-end wall-clock (prove GPU-bound with local caching).
 - Standardize the timing boundary (generate-only, with explicit device synchronization); discard warm-up iterations.
 - Re-benchmark on the target cluster hardware with batching, reporting measured throughput at multiple scales (1k / 2.5k / 10k) to expose memory fragmentation / thermal effects — not a single linear extrapolation.
 - Re-cost after D1 (gradient attribution) and D4 (dense captioning), both of which materially change the budget.
-

5. Phase 3 — Reproducibility, environment, and statistics hardening

- Local data cache (already implemented): the loader reads the locally-downloaded Parquet copy in preference to Hub streaming, keeping cluster runs GPU-bound and immune to the streaming `IncompleteRead` failure class.
 - Environment: containerize (Docker/Singularity) for the cluster; pin the full dependency set. Replace the fragile `transformers` version pin with a config-wrapper that injects the missing generation attributes, so the stack survives upstream updates. Document the exact install order (pinned-CUDA torch first) to avoid the CPU-torch replacement trap.
 - Precision: run forward-pass metrics in fp16 (fine); run gradient-based attribution (if D1 adopts it) in bf16 or fp32 and verify the attribution rankings don't change vs fp16.
 - Attention backend: document that SDPA on the cluster may route to FlashAttention, whose backward pass is non-deterministic — record baseline gradient matrices, and if strict cross-platform reproducibility of attribution is required, pin the attention backend.
 - Seeds: run the full study across multiple seeds (recommend ≥ 5) for sample selection and any stochastic step; report mean $\pm$ CI, not a single SEED=42 point.
 - Statistics: attach bootstrap CIs and Wilcoxon signed-rank tests to every headline metric; enforce the minimum-n floor before reporting a rate as stable.
 - Report generation: all tables/figures in the paper and memos are generated from the results CSVs at build time (no hand-copied numbers).
-

6. Phase 4 — Attribution stream(s)

- Primary: decoder→encoder cross-attention (as in the pilot), with the verified architecture handling (drop the global pooled token before reshaping the 576 image-patch tokens into the 24×24 grid; keep this behind an assertion and recompute the patch-grid arithmetic from the actual input resolution rather than hardcoding).
 - Second stream [D1]: Integrated Gradients on the vision encoder via Captum (already a project dependency), run in bf16/fp32. Report agreement between cross-attention and IG — this converts the attribution choice from a single fallback method into a corroborated result, which is what a top-tier venue expects.
 - Ranking use: feed cross-attention/IG mass as the importance signal for `region_ranking = attention` (Phase 1.1 tier 2), giving a defensible importance ranking rather than area.
 - Note for native-VQA models (Qwen2.5-VL): these are decoder-only, with no encoder-decoder cross-attention layer to read out. Attribution for them must use raw self-attention from selected layers/heads or a gradient method (IG) — plan for a model-specific attribution adapter, not a single shared method.
-

7. Phase 5 — Model and dataset scope [D2]

7.1 Models (three, with distinct jobs)

1. Florence-2-base-ft — keep as the instrumentable, region-native baseline. Runs the grounding-proxy pipeline.
2. Qwen2.5-VL (3B and 7B) — native-VQA + native grounding. Run two explicitly-labeled comparisons, because comparing a native-VQA model to Florence-2 on the same numbers is otherwise a confound:
 - Same-proxy comparison: constrain Qwen2.5-VL to the identical detect-then-threshold pipeline (it supports open-vocabulary grounding). Isolates "does a better detector make the existing proxy's explanations more faithful."
 - Native-VQA comparison: let Qwen2.5-VL answer each rule directly as free-text VQA, with its own attention/gradient attribution. Isolates "does end-to-end reasoning produce better explanations than a geometric proxy." This is the comparison that makes the metrics measure model faithfulness rather than proxy sensitivity.
 - 7B likely needs the cluster GPU and possibly quantization; 3B for faster iteration.
3. One frontier model (e.g. GPT-4o) — benchmark-only reference for answer accuracy. No gradient/attention attribution (API-only); include because reviewers in this space now expect a frontier comparison point.

7.2 Datasets (three, with distinct jobs)

1. ConstructionSite — scale from 163 to the full 3,004-image test split (train split available for supplementing minority classes). Pair with the multi-label fix (3.2) so `struck_by_risk` is powered rather than inherited at $n=13$.
2. SODA (published, ~19,846 images, 15 classes, multi-site/weather/viewpoint) — out-of-distribution robustness/completeness: does the explanation degrade gracefully on a different site distribution, not just on perturbed versions of the same images. Requires a class/rule mapping from SODA's taxonomy to the four safety rules.
3. CMA (published, ~1,595 clips, 7 action classes) — temporal explanation-stability: extend stability from "survives stochastic reruns on one static image" to "stays consistent frame-to-frame on the same hazard across a video clip." This is the genuinely novel contribution — no current construction-VLM work does explanation-stability over time. Requires frame sampling, per-frame grounding, and a frame-to-frame explanation-overlap metric.

7.3 Adversarial (Level-3) robustness [D5]

- Replace the 20-image synthetic-patch stretch test with a powered study: realistic hard-hat patches (not a yellow ellipse), pose-estimation-based placement (not a geometric head-box guess), and adequate n . Report spoofing rate only over candidates for which "fooling" is well-defined (baseline already "violation"), and report placement→flip correlation with sufficient power.

8. Recommended execution order and gating

1. Phase 1 (foundational fixes) — do first, behind flags, validate each gate. Cheap, unblocks everything.
2. Phase 0 decisions — resolve with Prof. Abdallah in parallel with Phase 1 (they gate Phases 4–5).
3. Phase 2 (metric validity) — after Phase 1 lands and ranking is validated.
4. Phase 3 (reproducibility/stats) — alongside Phase 2; required before any headline number is quoted.
5. Phase 4 (attribution) — after D1; feeds Phase 1.1's attention-based ranking and Phase 2's sparsity/drift.
6. Phase 5 (model/dataset scope) — after D2/D3/D4; the largest effort, gated on the foundations being correct so scale doesn't just produce a larger biased sample.

Global gate before scaling sample count: do not run the full 3,004-image study until Phase 1 fixes are validated. Scaling a biased pipeline only yields a larger, equally-biased result.

9. Deliverables checklist (per phase)

- Phase 1: flagged, validated fixes + before/after comparison on the 163-sample set showing the ranking gap narrowing and worker-loss correction applied.
- Phase 2: each metric emitting raw + corrected + size-invariant variants; text-ablation added to descriptive accuracy.
- Phase 3: containerized environment, multi-seed run harness, CI hardcoded-literal check, programmatic report generation.
- Phase 4: two attribution streams with an agreement report.
- Phase 5: three-model × three-dataset result matrix, with same-proxy and native-VQA comparisons clearly separated, and powered adversarial + temporal-stability studies.

10. Risk register

Risk	Likelihood	Mitigation
Rule-aware ranking changes recorded pilot numbers	High (intended)	Keep behind flag; freeze area default; report before/after explicitly
Dense captioning explodes decode cost	Medium	Measure on validation subset before adopting (D4); re-cost efficiency
Gradient attribution underflows in fp16	High if D1 adopts it	Run attribution in bf16/fp32; verify rankings unchanged
SODA/CMA taxonomy doesn't map cleanly to 4 rules	Medium	Build and review an explicit mapping before running; report unmapped classes
Cluster FlashAttention non-determinism perturbs attribution	Low-Medium	Pin attention backend for attribution runs; record baseline matrices
Native-VQA vs proxy comparison read as unfair	Medium	Always run and label both same-proxy and native-VQA comparisons
struck_by_risk still underpowered after multi-label	Low-Medium	Supplement from train split and/or CMA; enforce minimum-n before reporting